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Can we trust the model?

Week 12: Everything from the previous eleven weeks, pointed at a single question.

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By the end of this week you should be able to

Learning outcomes served: LO1, LO2, LO3, LO4, LO5

- Assess whether a reported performance figure supports the claim being made from it.
- Assess and interpret calibration, and distinguish it from discrimination.
- Quantify uncertainty in a performance estimate and report confidence intervals.
- Identify dataset shift and evaluate subgroup performance.
- Weigh computational and environmental cost, licensing and privacy alongside accuracy.
- Explain what external validation means and why internal validation cannot substitute for it.

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The biological problem

Part 1 · Why anyone cares about this week's method

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Biological context

- A model that is accurate on a retrospective dataset and useless in a clinic is a familiar object. The gap between the two is not usually caused by the algorithm. It is caused by evaluation that answered an easier question than the one that mattered.
- This week assembles the tools for asking the harder question.

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The method

Part 2 · Concepts, then the mathematics

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Discrimination versus calibration

Discrimination, measured by AUC, is whether the model ranks positives above negatives. Calibration is whether a stated probability of 0.8 corresponds to an 80 per cent event rate. A model can discriminate excellently and be badly calibrated, and for clinical decisions calibration is often the property that matters.

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Uncertainty in the estimate

On a 100-patient test set, an observed accuracy of 0.90 has a 95 per cent confidence interval of roughly 0.82 to 0.95. Two models reported as 0.90 and 0.93 on such a set are not distinguishable. Report intervals.

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Key concepts

Dataset shift Covariate shift is a change in the input distribution; label shift a change in the outcome base rate; concept shift a change in the relationship itself.

Subgroup performance Aggregate accuracy conceals systematic failure in a subgroup. Report performance by age, sex, ethnicity and site, and state the sample size in each cell so that the reader can see which estimates are unstable.

External validation Evaluation on data from a different source, collected independently. Internal cross-validation, however careful, cannot tell you whether the model transports.

Interpretability Coefficients, feature importances, SHAP values and attention maps describe the model. None of them establishes a mechanism in the biology, and treating them as if they did is the most common over-claim in the field.

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Key concepts (continued)

Computational and environmental cost Training large models has a measurable energy and carbon cost. It is a legitimate axis of model selection, and for a marginal accuracy gain it can be the deciding one.

Licensing and privacy A model licensed for research use only cannot be deployed clinically. Patient data carries obligations that survive de-identification. Both are constraints on the project, not paperwork at the end of it.

Clinical validation Prospective evaluation of whether using the model changes patient outcomes. It is a different kind of study from anything in this module, and it is the one that decides whether the work mattered.

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Uncertainty in a performance estimate

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```
Wilson 95% CI  around  p̂ = 0.90,  n = 100   →   [0.824, 0.945]
```

An observed accuracy is an estimate with a standard error. On a 100-patient test set, models reported as 0.90 and 0.93 are not distinguishable, and reporting the difference as real is a common error.

- **p̂** the observed proportion correct
- **n** test-set size; the interval narrows as √n
- **Wilson** preferred to the normal approximation, which misbehaves near 0 and 1
- **report** the interval, not the point estimate

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Calibration versus discrimination

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$$\text{Calibrated:} \quad P(y = 1 \mid \hat{p} = q) = q \quad \text{for all } q$$

Discrimination, measured by AUC, is whether the model ranks positives above negatives. Calibration is whether a stated probability means what it says. AUC depends only on the ranking, so it is completely blind to calibration.

AUC can be excellent while calibration is poor

matters when the probability itself informs a decision

fixes Platt scaling (a logistic fit) or isotonic regression

check a reliability diagram, binned by predicted probability

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Interrogating a 97% accuracy claim

None of these is hostile. A good paper answers them without being asked.

Question	Why it matters
How many test samples?	97% on n=150 has a 95% CI of about 92.7 to 98.8%
What is the class balance?	At 8% positive, predicting 'negative' always scores 92%
How were the data split?	Tiles from one patient on both sides inflate by an unknown amount
Was there external validation?	One scanner, one protocol, one population cannot be detected from inside
Is it calibrated?	A model can rank well and still state the wrong probability
Who is in the data?	Aggregate accuracy conceals systematic subgroup failure
Sensitivity and specificity separately?	A single figure hides which error the model prefers to make

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Dataset shift

A model deployed at a different hospital typically meets all three at once.

Kind	What changes	Biological example
Covariate shift	P(x)	A different scanner or assay platform
Label shift	P(y)	Deployment in a lower-prevalence population
Concept shift	P(y | x)	Treatment guidelines change, so the same features imply a different outcome

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The trust checklist

A high-performing model is not necessarily a scientifically valid or clinically useful one.

- Report an interval, not a point estimate
- Report the baseline alongside every result
- Assess calibration, not only discrimination
- Report subgroup performance with the sample size in each cell
- State what external validation was done, or that none was
- State the computational and environmental cost
- Check the licence: research-only weights cannot be deployed clinically
- For foundation models: was your test set in the pretraining corpus? Can you archive the weights you depended on?

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Worked example

Part 3 · Interrogating a 97 per cent accuracy claim

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Interrogating a 97 per cent accuracy claim

- A paper reports 97 per cent accuracy for cancer detection. Take it apart.
- How many samples? On 200 samples, 97 per cent has a 95 per cent confidence interval of roughly 93.5 to 98.7 per cent. On 50 samples it is roughly 88 to 99.5 per cent, which is compatible with a substantially worse model.
- What is the class balance? If 95 per cent of samples are negative, then always predicting negative scores 95 per cent, and 97 per cent is an improvement of two points over doing nothing at all.
- What was the protocol? Was any preprocessing, feature selection or hyperparameter choice made using the test data? Were repeat samples from the same patient split across folds? Recall from Week 3 that you cannot detect this from the score.

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Interrogating a 97 per cent accuracy claim (continued)

- Was there external validation? A model validated only on its development cohort has been shown to fit that cohort. That is a weaker claim than it appears.
- Is it calibrated? For a screening application, a model that ranks well but systematically overstates its confidence will produce the wrong number of referrals.
- Who is in the data? If the cohort is 90 per cent one demographic group, the reported figure describes that group. Performance elsewhere is unmeasured, not equal.
- What are the sensitivity and specificity, separately? A single accuracy figure conceals which error the model prefers to make, and in medicine those two errors are almost never equally costly.
- None of these questions is hostile. They are the questions the authors should have answered, and a good paper answers them without being asked.

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A high-performing model is not necessarily a scientifically valid or clinically useful one.

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Try this interactively

Would You Trust This Model?

What questions must a performance claim survive?

`weeks/week12.html` · `Interactive exploration`

Open the week page on the course website and scroll to Interactive exploration.

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Try this interactively

Should I Use a Foundation Model?

Should I use a foundation model for this problem at all?

`weeks/week12.html` · `Interactive exploration`

Open the week page on the course website and scroll to Interactive exploration.

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The foundation-model connection

Part 4 · The extra questions foundation models raise

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The extra questions foundation models raise

- Everything above still applies, and five things are added. Was your test set in the pretraining corpus? What is in that corpus, and which populations or organisms does it under-represent? What is the licence on the weights, and does it permit your use?
- What does inference cost at the scale you intend to deploy? And can you reproduce the result at all, given that the pretrained weights are a dependency you do not control and may not be able to archive?
- None of these has a comfortable general answer, which is exactly why they have to be asked case by case.

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Interpretation and practical

Part 5 · What the result means, and what you will do with it

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Take one result you produced in this module. Write the paragraph a reviewer would write recommending rejection. Then decide, honestly, whether you can answer it.

Think like a scientist. Bring your answer to the laboratory.

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Auditing a model you did not build

This week's laboratory · Apply the full trust checklist to a model, including one of your own. · 2 hours

- Take your best model from Week 10 or 11. Compute a bootstrap 95 per cent confidence interval for its accuracy. Report the interval, not the point estimate.
- Plot a calibration curve with sklearn.calibration.calibration_curve. Is it calibrated? Apply Platt scaling and isotonic regression and replot.
- Simulate covariate shift: scale one informative feature in the test set by 1.3, as a different scanner might. Report the accuracy drop.
- Split the test set by a proxy subgroup variable of your choice and report performance in each subgroup with sample sizes. Identify which estimates are too small to interpret.
- Estimate the compute cost of training your model and of the largest model discussed in this module. Comment on the ratio.
- Complete the trust checklist from the lecture for your model. Be honest about the entries you cannot satisfy.
- Finally, exchange notebooks with another student and audit theirs. Write the three questions you would ask before believing their headline number.

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Self-check

- A model has AUC 0.94 but its predicted probabilities are systematically too high. What is wrong, and does it matter?
- A) Nothing is wrong, AUC is what matters
- B) It discriminates well but is poorly calibrated, which matters whenever the probability itself is used to make a decision
- C) The AUC must be wrong
- D) It needs more training data

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Self-check

- Two models score 0.90 and 0.93 accuracy on the same 100-sample test set. What can you conclude?
- A) The second is better
- B) The confidence intervals overlap substantially, so the difference is not established by this evidence
- C) The first is overfitting
- D) You need a third model

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References and further reading

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Questions

That is the module. Both projects are described in full on the assessment page.

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